Structural certificates for M_{12} and $M_{12} : 2$

Private L054 finite application

Theorem 1. *All 147 rows of the current M_{12} companion table and all 213 rows of its $M_{12} : 2$ table have the displayed structural descriptions, relative to the explicitly acquired numerical and named-model interfaces. Each description asserts existence of its displayed decomposition, without asserting that the label determines a unique isomorphism type.*

Exact source and acquired groups. These current companion PDFs are attributed to Thomas Connor and Dimitri Leemans. All fifteen native pages were directly read. We acquire the accepted complete numerical inventories, source-row bijections and literal embedded subgroups. Every group expression in both imported arrays is a literal `Group` of permutations; no named constructor is substituted inside a proper row. Their degrees are 12 and 396. The original successful `MathieuGroup(12)` execution, its exact driver, clean receipt and recorded input/output hashes are acquired. Its top generators equal the current twelve-point top generators. GAP Reference Manual 50.1-13 supplies the named Mathieu constructor interface explicitly; it is not inferred from group order. Earlier concrete M_9, M_{10}, M_{11} and other structural models are acquired with their complete accepted proofs.

The 2015 article's left-association convention is retained. In particular, $3 : S_3 \cdot 2 \cdot 2$ in M_{12} rows 38 and 40 has two dots, whereas $2^{1+4} \cdot 2^2$ in $M_{12} : 2$ row 30 has an elementary abelian quotient of order four. We preserve $4^2 = C_4^2$, unsigned extraspecial atoms and the acquired dicyclic Q_{12} convention. There is no assertion that these current PDF bytes are a historical edition.

Lemma 2 (Literal faithful right-coset transport)**.** *Suppose a finite permutation group G, its order n, an actual subgroup H, and other actual subgroups $K_j \leq G$ are acquired. Let $|H| = h$ be verified by literal closure and set $k = n/h$. Suppose k permutations t_i, starting with identity, are verified words in the ordered generators of G, and $t_i t_j^{-1} \notin H$ for distinct i, j. For each generator x of G, let a permutation p_x of $\{1, \ldots, k\}$ satisfy*

$$t_i x t_{p_x(i)}^{-1} \in H \quad \text{for all } i.$$

These permutations induce the actual coset homomorphism. If their literal closure has order n, that homomorphism is faithful. The same equations for each generator of each K_j identify all their images exactly.

Proof. The words prove $t_i \in G$. The distinct right cosets Ht_i exhaust G by the acquired index. The displayed membership is exactly $Ht_i x = Ht_{p_x(i)}$, hence gives right multiplication on this complete coset set. With right-action permutation composition this is a homomorphism. Its image order equals $|G|$, so the kernel is trivial. Since each further generator lies in the acquired G, its equations determine its unique image under this same map. Thus the actual subgroup embeddings are preserved. Equal orders are used only after constructing the homomorphism. □

The actual compact model. For the degree-396 group we use the retained subgroup at source row 3, of order 7,920. Its M_{11} name is unnecessary for the coset argument. The acquired ambient order is 190,080, so the index is 24. Candidate breadth-first words supply representatives. The independent literal reader checks their words, 276 pairwise coset distinctions and 11,112 coset equations for the ambient generators and all 213 subgroup generator lists. The image closes to 190,080 elements. All later certificates therefore operate on exact faithful 24-point images, not unrelated groups of matching orders. The original 396-point arrays and source assignments remain bound. No ambient subgroup enumeration or new automorphism computation is used.

Lemma 3 (Nonsplitting with Klein four quotient)**.** *Let $N \lhd K$, and let $x, y \in K$ satisfy $x^2, y^2, [x, y] \in N$, with N, Nx, Ny, Nxy disjoint and covering K. Then $K/N \cong C_2^2$, and the extension splits if and only if there are $a \in Nx$, $b \in Ny$ with $a^2 = b^2 = 1$ and $ab = ba$.*

Proof. The relations and four distinct normal forms give the stated quotient. A complement supplies the indicated lifts. Conversely, such lifts define a homomorphism from C_2^2 into K; its composite with the quotient map is the prescribed isomorphism. It is therefore injective, and its image is a complement. Checking every pair in the two cosets proves nonsplitting when none satisfies the three relations. No failed complement search is a premise. □

The actual structural certificate. The registry contains 128 acquired concrete models and one new retained Mathieu constructor input. There are 23 new decompositions: eleven split, three direct and nine nonsplit. Each factor/occurrence map is an explicit identity-preserving bijection checked on all right-generator edges. The acquired MC.3 word-induction theorem proves it an isomorphism. Normality is checked on subgroup and ambient generators. A complement has trivial intersection with the normal subgroup and the required product of orders; a direct product additionally has commuting factors. A displayed nontrivial semidirect action has a noncommuting generator pair.

Six new prime-index nonsplit extensions use 152 checks of all outside elements. The three composite cases use the acquired cyclic-six and S_3 lift criteria, and the preceding Klein four lemma: respectively 32, 1,024 and 1,024 lift tests. In every case quotient relations and all quotient cosets are checked first. In particular, the new criterion pays exactly $2^{1+4} \cdot 2^2$, not a cyclic quotient substitute. Every final map lands in the exact source-row subgroup (or its proved faithful image). The 397 maps and 1,368,412 right-generator equations establish all 360 asserted structures.

Execution and independently checked interfaces. The candidate coset pass completed in 1.54 seconds. The sole bounded structural candidate pass completed all 360 rows in 3.77 seconds. GAP's normal-subgroup, complement and isomorphism operations supplied witnesses only. The replay uses literal permutations and all retained certificates; it repeats neither discovery nor subgroup classification.

An initial compact-output reader rejected a physical GAP line continuation between a digit and a comma before any mathematical check. The raw output and old reader are preserved. The corrected reader removes only physical backslash–newline at numeric array-token boundaries and then parses strict JSON; permutation domains, representative words and all equations are checked. No candidate output was edited.

The independent small action controls enumerate S_4 literally. Its nonnormal point stabilizer verifies the right-coset convention and rejects inverse-action substitution. Its normal Klein four subgroup supplies a nonfaithful action which fails the final image-order test. Other damages include repeated cosets, incorrect words, Boolean word indices, changed row images, and nonpermutation images. A changed actual 396-point row image also fails its coset equations. Independent $D_8/Z(D_8)$ and C_2^3/C_2 controls distinguish nonsplit and split Klein four extensions; dependent quotient generators fail the four-coset check.

Full-certificate controls change identity or a right-generator edge, omit an element or source row, alter source-row assignments, actual generators, source powers and dots, the named Mathieu input, complements, factor references and an actual Klein four lift. Ordinary and optimized Python must produce identical reports. The source syntax is joined to the actual retained embedded rows, not merely to their numerical orders.

Scope and attribution. The result adds 360 structural row joins for the two current tables. Their complete numerical classifications, named constructors and earlier registry proofs are explicit retained inputs. The finite source-sensitive application and the new literal coset and Klein-quotient checks are authored here; root directed compact-action and registry reuse. These elementary arguments carry no novelty claim. There is no new ambient census, historical-byte identity, uniquely determined extension name, whole-source admission, question resolution, formalization or measurement credit.